

Equations

Week 3D

Overview

Inequalities

Key Ideas - Introduction to Inequalities

- Properties of Inequalities

- Interval Notation

- Solving Inequalities

- Solving Absolute Value Inequalities

Key Idea – Introduction to Inequalities

An inequality is a mathematical expression involving one of the symbols

$$\geq, \leq, < \text{ or } >$$

In this section you will learn how to correctly manipulate inequalities and to solve inequalities algebraically.

The meaning of each symbol:

- $>$ means *“is greater than”*
- $<$ means *“is less than”*
- $\geq$ means *“is greater than or equal to”*
- $\leq$ means *“is less than or equal to”*

Given two real numbers a, b , there are **3 possible orderings** of the two numbers on the number line:

$$a = b$$

$$a < b$$

$$a > b$$

Consider two real numbers a and b . In relation to the number line, the three statements below have the following meaning:

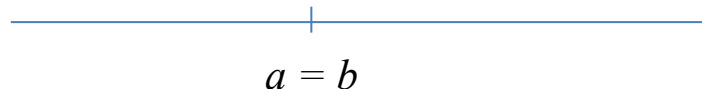
- $a < b$ means that the **number a lies to the left of the number b** .

We say that a is **less than b** .



- $a = b$ means that the **numbers a and b occupy the same position**.

We say that a is **equal to b** .



- $a > b$ means that the **number a lies to the right of the number b** .

We say that a is **greater than b** .



Key Idea – Properties of Inequalities

Below is a summary of the results of the above investigations.

Trichotomy Property:

For any two real numbers a and b , exactly one of the following is true:

$$a < b, a = b, a > b$$

Transitive Properties of Inequality:

If $a < b$ and $b < c$, then $a < c$

If $a > b$ and $b > c$, then $a > c$

Addition and Subtraction of Inequalities

If a number is added to (or subtracted from) both sides of the inequality, the sign of the inequality **does not change direction**.

$$\text{If } a > b, \text{ then } a \pm c > b \pm c$$

Multiplication and Division of an Inequality by a Positive Number

If both sides of an inequality are multiplied by a **positive number**, the inequality **sign does not change direction**.

$$\text{If } a > b \text{ and } c > 0 \text{ then } ac > bc$$

Multiplication and Division of an Inequality by a Negative Number

If both sides of an inequality are multiplied by a **negative number**, the direction of the **inequality sign is reversed**.

$$\text{If } a > b, \text{ and } c < 0, \text{ then } ac < bc$$

Key Idea – Interval Notation

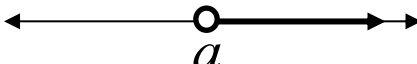
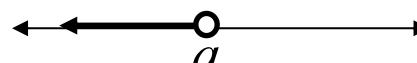
Interval notation is a common way to express the solution set of an inequality. It is important notation because we express solution sets in calculus this way.

The solution set of an inequality can also be displayed on the **number line**. When graphing an interval on the number line, draw a thick line to show the values that are included.

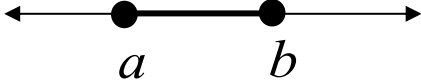
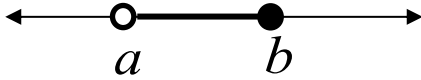
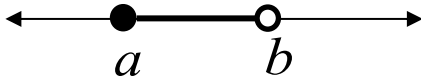
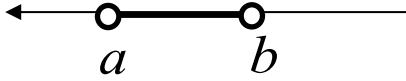
Also:

- Use a **filled-in circle** or **square bracket** when you want to include an end value.
- Use an **open circle** or **round bracket** when you don't want to include an end value.

If $x, a \in \mathbb{R}$ then:

Inequality	Interval Notation	Set Notation	Graph	Description
$x \geq a$	$[a, \infty)$	$\{x : x \in \mathbb{R} \text{ and } x \geq a\}$		x is greater than or equal to a
$x \leq a$	$(-\infty, a]$	$\{x : x \in \mathbb{R} \text{ and } x \leq a\}$		x is less than or equal to a
$x > a$	(a, ∞)	$\{x : x \in \mathbb{R} \text{ and } x > a\}$		x is greater than a
$x < a$	$(-\infty, a)$	$\{x : x \in \mathbb{R} \text{ and } x < a\}$		x is less than a

If $a, b \in \mathbb{R}$ and $a < b$, then:

Inequality	Interval Notation	Set Notation	Graph	Description
$a \leq x \leq b$	$[a, b]$	$\{x : x \in \mathbb{R} \text{ and } a \leq x \leq b\}$		The set is called a closed interval . x is between a and b , including a and b
$a < x \leq b$	$(a, b]$	$\{x : x \in \mathbb{R} \text{ and } a < x \leq b\}$		The set is called a half-open interval . x is between a and b , including b
$a \leq x < b$	$[a, b)$	$\{x : x \in \mathbb{R} \text{ and } a \leq x < b\}$		The set is called a half-open interval . x is between a and b , including a
$a < x < b$	(a, b)	$\{x : x \in \mathbb{R} \text{ and } a < x < b\}$		the set is called a open interval . x is between a and b , not including a or b

Key Idea – Solving Inequalities

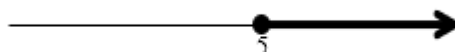
To solve an inequality we use the **same methods** as for solving an equation, but we must also **follow the rules for inequalities**.

Examples: Solve the following inequalities and graph each solution on a number line:

$$2x + 5 \geq 15$$

$$2x \geq 10$$

$$x \geq 5$$

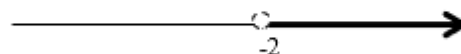


In interval notation $[5, \infty)$

$$6 > -3x$$

$$-3x < 6$$

$$x > -2$$

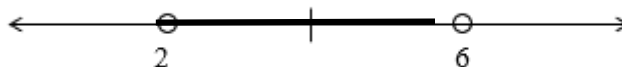


In interval notation $(-2, \infty)$

$$4 < 2x < 12$$

$$\frac{4}{2} < \frac{2x}{2} < \frac{12}{2}$$

$$2 < x < 6$$



In interval notation $(2, 6)$

Exercise: Solve the following inequalities:

(a) $2(x - 3) \leq 3(x - 1)$

$$2x - 6 \leq 3x - 3$$

$$(-3x)$$

$$-x - 6 \leq -3$$

$$(+6)$$

$$-x \leq 3$$

$$(\div -1)$$

$$x \geq -3$$

in interval notation: $[-3, \infty)$

$$(b) \frac{2}{3} < 2x + 3 \leq 15$$

(-3)

$$\frac{2}{3} - 3 < 2x + 3 - 3 \leq 15 - 3$$

$$-\frac{7}{3} < 2x \leq 12$$

$(\div 2)$

$$-\frac{7}{6} < x \leq 6$$

in interval notation: $\left(-\frac{7}{6}, 6\right]$

$$(c) \frac{x}{6} + \frac{2x+5}{3} < 4$$

$$6 \left(\frac{x}{6} + \frac{2x+5}{3} \right) < 6 \times 4 \quad (\times 6)$$

$$\frac{6x}{6} + \frac{6(2x+5)}{3} < 24$$

$$x + 2(2x+5) < 24$$

$$x + 4x + 10 < 24$$

$$5x + 10 < 24 \quad (-10)$$

$$5x < 14 \quad (\div 5)$$

$$x < \frac{14}{5}$$

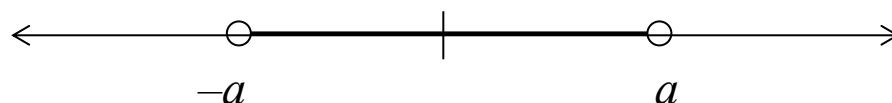
in interval notation: $(-\infty, \frac{14}{5})$

Key Idea – Solving Absolute Value Equations with Inequalities

On the number line, the relationships between inequalities and absolute value are as follows:

- $|x| < a$, means the distance of x from zero (the origin) is **less than a units**.

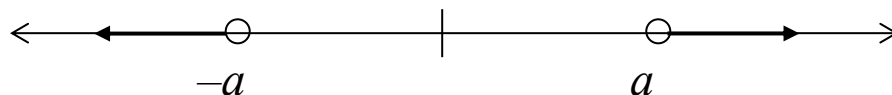
This inequality can be written as $-a < x < a$.



This inequality can also be written in the form $(-a, a)$.

- $|x| > a$, means the distance of x from zero (the origin) is **greater than a units**.

This inequality can be written as $x < -a$ or $x > a$, where both of these inequalities are required.



This inequality can also be written in the form $(-\infty, -a) \cup (a, \infty)$.

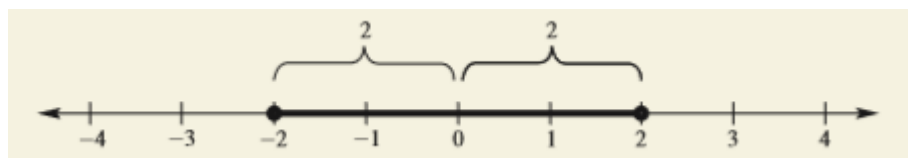
Note: Similar relationships hold for the inequalities $\geq$ and $\leq$.

Example: Solve $|x| \leq 2$

$|x| \leq 2$ means the distance of x from zero is less than or equal to 2 units (in either direction).

Notice that there is one region on the number line. We can write this as a single statement:

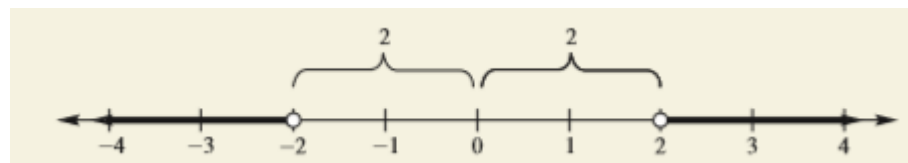
$$-2 \leq x \leq 2.$$



Example: $|x| > 2$

$|x| > 2$ means the distance of x from zero is greater than 2 units (in either direction).

Notice that the solution involves two separate regions on the number line, so the solution requires two separate inequalities $x < -2$ or $x > 2$.



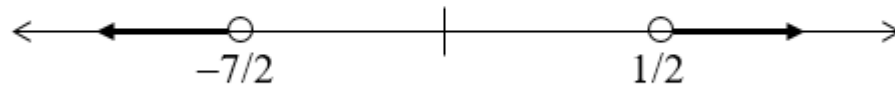
Example: Solve $|2x + 3| > 4$ and graph the solution on the number line:

$$2x + 3 < -4 \qquad 2x + 3 > 4$$

$$2x < -7 \qquad \text{or} \qquad 2x > 1$$

$$x < -\frac{7}{2} \qquad x > \frac{1}{2}$$

Solution on number line



Exercise: Solve the following absolute value inequalities:

(a) $|3 - 8x| > 2$

$$3 - 8x > 2 \quad (-3) \quad \text{or}$$

$$-8x > -1 \quad (\div -8)$$

$$x < \frac{1}{8}$$

$$3 - 8x < -2 \quad (-3)$$

$$-8x < -5 \quad (\div -8)$$

$$x > \frac{5}{8}$$

(b) $\left| \frac{3x}{2} - 1 \right| < 1$

$$-1 < \frac{3x}{2} - 1 < 1 \quad (\times 2)$$

$$-2 < 3x - 2 < 2 \quad (+2)$$

$$0 < 3x < 4 \quad (\div 3)$$

$$0 < x < \frac{4}{3}$$